S181M116 Derivatives and Alternative Investments: Mathematical Foundations

Swapnil Singh

This note reviews the mathematical foundations needed for the course. It is not a general mathematics textbook. It is a practical toolkit for derivatives and alternative investments: algebra, discounting, payoff functions, no-arbitrage logic, probability, normal and lognormal distributions, calculus, Taylor approximations, optimization, portfolio variance, regression beta, binomial trees, and the first ideas behind Brownian motion and Itô's lemma. The note is example-heavy because the course is mathematically intensive and the formulas are useful only when students can interpret them economically.

> **How to use this note.** Read Sections 1–6 before the first class if the notation is unfamiliar. Return to Sections 7–12 before the lectures on hedging, binomial trees, Black–Scholes–Merton, Greeks, volatility, and real options. The goal is working fluency: you should be able to compute, draw, explain, and check each object.

Why Mathematics Matters in This Course

Derivatives are contracts whose cash flows depend on future variables. This means that almost every important question has a mathematical form:

$$\text{contract value today} = \text{value of uncertain future cash flows.}$$

The course repeatedly uses four ideas.

1. **Payoffs are functions.** A forward payoff, option payoff, swap payment, and carried-interest payoff are all functions of future variables.
2. **Time matters.** A euro today and a euro next year are not the same object. Discounting converts future cash flows into present values.
3. **Risk matters.** Future cash flows are random. We need probability, expectation, variance, covariance, and distributional thinking.
4. **Replication matters.** If two strategies generate the same future cash flows, they should have the same current value. This is the no-arbitrage principle.

Central habit. Do not memorize a formula without asking three questions: What is being measured? What are the units? Under what assumptions is the formula valid?

A Map of the Toolkit

Course topic	Mathematical tools used
Forwards and futures	Algebra, compounding, discounting, no-arbitrage replication, linear payoff functions
Hedging with futures	Covariance, correlation, regression beta, variance minimization
Interest rates and swaps	Present value, discount factors, zero rates, forward rates, sums of cash flows
Options	Positive-part functions, convexity, inequalities, put-call parity, piecewise functions
Binomial trees	Weighted averages, replication, backward induction, risk-neutral probabilities
Black–Scholes–Merton	Calculus, partial derivatives, Taylor expansion, Brownian motion, Itô's lemma
Greeks and risk	First and second derivatives, scenario approximation, portfolio aggregation
Alternative investments	Net present value, IRR, option-like flexibility, discounted cash-flow logic

Notation and Algebra

Variables, Parameters, and Subscripts

A *variable* can change. A *parameter* is treated as fixed within the calculation. In derivatives notation:

$$S_0 = \text{spot price today}, \qquad S_T = \text{spot price at maturity},$$

$$K = \text{strike or delivery price}, \qquad T = \text{time to maturity}.$$

The subscript 0 usually means today. The subscript T means at maturity.

Example 1 (Reading notation)
If a European call has payoff

$$(S_T - K)^+,$$

then S_T is random before maturity, while K is fixed by the option contract. The plus sign does not mean "add something." It means positive part:

$$x^+ = \max(x, 0).$$

Units

Financial formulas are easier when units are explicit. If S_T and K are both in dollars per share, then $S_T - K$ is also dollars per share. If the contract is for N shares, the total payoff is

$$N(S_T - K).$$

Example 2 (Units in a forward payoff)
A forward contract is written on 5,000 barrels of oil. The delivery price is $K = 74$ dollars per barrel. At maturity the spot price is $S_T = 79$ dollars per barrel. The long forward payoff per barrel is

$$S_T - K = 79 - 74 = 5.$$

The total payoff is

$$5{,}000 \times 5 = \$25{,}000.$$

The unit check is

$$\textit{barrels} \times \frac{\textit{dollars}}{\textit{barrel}} = \textit{dollars}.$$

Fractions, Percentages, and Basis Points

A percentage rate must be converted into decimal form before it is used in a formula:

$$5\% = 0.05, \qquad 12.5\% = 0.125.$$

One basis point is one-hundredth of one percentage point:

$$1 \text{ bp} = 0.01\% = 0.0001.$$

Therefore 75 basis points is 0.75% or 0.0075.

Example 3 (Basis points)
An interest rate increases from 3.20% *to* 3.65%*. The change is*

$$3.65\% - 3.20\% = 0.45\% = 45 \textit{ bp}.$$

In decimal form the rate increases from 0.0320 *to* 0.0365*.*

Powers and Exponents

The expression x^n means x multiplied by itself n times. The expression e^x uses the exponential function, where $e \approx 2.71828$. Exponentials are central because continuously compounded interest grows as e^{rT}.

Useful rules:

$$e^a e^b = e^{a+b}, \qquad \frac{e^a}{e^b} = e^{a-b}, \qquad (e^a)^b = e^{ab}.$$

For any positive number A,

$$\ln(A)$$

is the natural logarithm, the inverse of the exponential:

$$\ln(e^x) = x, \qquad e^{\ln(A)} = A.$$

Example 4 (Solving with logs)
Suppose a price grows from 100 to 110 over one year under continuous compounding. Find the continuously compounded return r:

$$110 = 100e^r.$$

Divide by 100:

$$1.10 = e^r.$$

Take logs:

$$r = \ln(1.10) \approx 0.0953.$$

The continuously compounded return is 9.53%.

Summation Notation

The symbol $\sum$ means add terms:

$$\sum_{t=1}^{T} CF_t = CF_1 + CF_2 + \cdots + CF_T.$$

For discounted cash flows,

$$\text{PV} = \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t}.$$

Example 5 (Present value as a sum)
A project pays 100 at the end of year 1, 120 at the end of year 2, and 150 at the end of year 3. If the annual discount rate is 5%, then

$$\text{PV} = \frac{100}{1.05} + \frac{120}{1.05^2} + \frac{150}{1.05^3} \approx 333.18.$$

Time Value of Money

Future Value and Present Value

If money earns interest, one unit today grows into more than one unit tomorrow. With annual compounding at rate r,

$$\text{FV} = \text{PV}(1+r)^T.$$

Solving for present value gives

$$\text{PV} = \frac{\text{FV}}{(1+r)^T}.$$

Definition 1 (Discount factor)
The discount factor $D(0,T)$ is the price today of one unit of currency paid at time T with no default risk. Under annual compounding,

$$D(0,T) = \frac{1}{(1+r)^T}.$$

Under continuous compounding,

$$D(0,T) = e^{-rT}.$$

Example 6 (Discounting one cash flow)
You will receive 1,000 in two years. The annual interest rate is 4%. The present value is

$$\text{PV} = \frac{1{,}000}{1.04^2} \approx 924.56.$$

Under continuous compounding at 4%, the present value is

$$\text{PV} = 1{,}000e^{-0.04\times 2} \approx 923.12.$$

Compounding Conventions

With m compounding periods per year, the future value of one unit invested for T years at nominal annual rate r is

$$\left(1+\frac{r}{m}\right)^{mT}.$$

As m becomes very large, this approaches continuous compounding:

$$\lim_{m\to\infty}\left(1+\frac{r}{m}\right)^{mT} = e^{rT}.$$

Example 7 (Annual, quarterly, and continuous compounding)
Invest 100 for two years at a quoted annual rate of 6%.

$$\begin{aligned}
\textit{Annual compounding:}\quad & 100(1.06)^2 = 112.36,\\
\textit{Quarterly compounding:}\quad & 100\left(1+\frac{0.06}{4}\right)^8 \approx 112.65,\\
\textit{Continuous compounding:}\quad & 100e^{0.06\times 2} \approx 112.75.
\end{aligned}$$

More frequent compounding produces slightly more interest for the same nominal rate.

Converting Rates

If R_c is a continuously compounded rate and R_m is a rate compounded m times per year, the same one-year accumulation requires

$$e^{R_c} = \left(1+\frac{R_m}{m}\right)^m.$$

Therefore

$$R_c = m\ln\left(1+\frac{R_m}{m}\right), \qquad R_m = m\left(e^{R_c/m}-1\right).$$

Example 8 (Convert an annual rate to continuous compounding)
A bank quotes 8% with annual compounding. The equivalent continuously compounded rate is

$$R_c = \ln(1.08) \approx 0.07696.$$

So 8% annual compounding is equivalent to about 7.696% continuous compounding.

Zero Rates, Discount Factors, and Forward Rates

A zero-coupon bond pays one cash flow at maturity. If the continuously compounded zero rate for maturity T is $R(0,T)$, then the discount factor is

$$P(0,T) = e^{-R(0,T)T}.$$

If we know discount factors, we can recover zero rates:

$$R(0,T) = -\frac{1}{T}\ln P(0,T).$$

The forward rate from T_1 to T_2 is the rate agreed today for borrowing or lending between those future dates. With continuous compounding,

$$f(T_1,T_2) = \frac{R(0,T_2)T_2 - R(0,T_1)T_1}{T_2 - T_1}.$$

Example 9 (Forward rate from zero rates)
The one-year continuously compounded zero rate is 3% and the two-year zero rate is 4%. The one-year forward rate starting one year from now is

$$f(1,2) = \frac{0.04 \times 2 - 0.03 \times 1}{2-1} = 0.05.$$

The implied forward rate is 5%.

Net Present Value

Net present value subtracts the initial cost from the present value of future cash flows:

$$\text{NPV} = -I_0 + \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t}.$$

The financial interpretation is simple: if NPV > 0, the project is worth more than it costs using discount rate r.

Example 10 (Project NPV)
A real asset investment costs 500 today and is expected to generate 180 at the end of each of the next three years. At a discount rate of 7%,

$$\text{NPV} = -500 + \frac{180}{1.07} + \frac{180}{1.07^2} + \frac{180}{1.07^3} \approx -27.63.$$

At this discount rate, the project destroys value unless it has additional strategic value, such as an option to expand later.

Functions and Payoffs

Functions

A function maps an input into an output. In finance, the input is often a future price and the output is a payoff:

$$\text{payoff} = g(S_T).$$

The same function can be represented by a formula, a table, or a graph.

Example 11 (A linear payoff function)
A long forward with delivery price $K = 100$ has payoff

$$g(S_T) = S_T - 100.$$

S_T	70	90	100	130
$g(S_T)$	-30	-10	0	30

The payoff increases one-for-one with the terminal spot price.

Positive-Part Notation

The positive part of x is

$$x^+ = \max(x, 0) = \begin{cases} x, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

This notation is used constantly for options:

$$\text{Call payoff} = (S_T - K)^+, \qquad \text{Put payoff} = (K - S_T)^+.$$

Example 12 (Call option payoff)
A call has strike $K = 50$. Its payoff is

$$(S_T - 50)^+.$$

S_T	35	45	50	60	80
Call payoff	0	0	0	10	30

The option is valuable at maturity only when the underlying price exceeds the strike.

Example 13 (Put option payoff)
A put has strike $K = 50$. Its payoff is

$$(50 - S_T)^+.$$

S_T	35	45	50	60	80
Put payoff	15	5	0	0	0

The put is valuable at maturity when the underlying price is below the strike.

Payoff Versus Profit

Payoff is the cash flow at maturity. Profit subtracts the initial price of the position, usually adjusted for financing. Ignoring financing, a call bought for premium c_0 has profit

$$(S_T - K)^+ - c_0.$$

Example 14 (Option payoff and profit)
A trader buys a call with strike 100 for premium 7. Ignoring interest on the premium:

$$\textit{profit} = (S_T - 100)^+ - 7.$$

S_T	80	95	100	107	120
Payoff	0	0	0	7	20
Profit	-7	-7	-7	0	13

The break-even terminal price is 107.

Piecewise Functions

Many derivative payoffs are piecewise: the formula changes depending on the state. A call payoff can be written as

$$(S_T - K)^+ = \begin{cases} 0, & S_T \le K, \\ S_T - K, & S_T > K. \end{cases}$$

Example 15 (Covered call payoff)
A covered call consists of buying one share and selling one call with strike K. Ignoring the option premium, the terminal payoff is

$$S_T - (S_T - K)^+.$$

If $S_T \le K$, the call expires worthless and the payoff is S_T. If $S_T > K$, the short call loses $S_T - K$, so the total payoff is

$$S_T - (S_T - K) = K.$$

Thus

$$S_T - (S_T - K)^+ = \begin{cases} S_T, & S_T \le K, \\ K, & S_T > K. \end{cases}$$

The covered call keeps downside stock exposure but caps the upside.

Convexity

A function is convex when the line segment between two points on the graph lies above the graph. Intuitively, a convex payoff benefits from movement in either direction. Options have convex payoffs.

For a differentiable function, convexity corresponds to a nonnegative second derivative:

$$g''(x) \geq 0.$$

Linear functions are both convex and concave because their second derivative is zero.

Example 16 (Why a call is convex)
The call payoff is flat when $S_T < K$ and then rises with slope one when $S_T > K$. The slope jumps upward from 0 to 1 at K. That upward change in slope is the source of convexity.

> **Financial intuition.** Forwards give linear exposure. Options give nonlinear exposure. Nonlinearity is why option pricing requires more machinery than forward pricing.

No-Arbitrage Logic

Law of One Price

The law of one price says that two assets with exactly the same future cash flows in every state should have the same price today. If not, a trader can buy the cheap asset and sell the expensive asset.

Definition 2 (Arbitrage)
An arbitrage is a trading strategy with no initial cost or a negative initial cost, no possibility of future loss, and a positive probability of future gain.

Most textbook pricing arguments use a stronger practical version: if two self-financing strategies have the same future payoff, their current costs must be equal.

Forward Price for a Non-Dividend-Paying Asset

Consider a stock with current price S_0 and no dividends before time T. Compare two strategies:

1. Buy the stock today and finance the purchase by borrowing S_0.
2. Enter a forward contract to buy the stock at time T for delivery price F_0.

At time T, both strategies produce ownership of one share. The first strategy requires repayment $S_0 e^{rT}$. The second strategy requires payment F_0. No-arbitrage implies

$$F_0 = S_0 e^{rT}.$$

Example 17 (Cash-and-carry arbitrage)
Suppose $S_0 = 100$, $r = 5\%$, $T = 1$, *and the market forward price is 108. The no-arbitrage forward price is*

$$100e^{0.05} \approx 105.13.$$

The forward is too expensive. A trader can:

1. *borrow 100,*
2. *buy the stock,*
3. *short the forward at 108.*

At maturity, deliver the stock into the forward, receive 108, and repay $100e^{0.05} = 105.13$. *The locked-in profit is*

$$108 - 105.13 = 2.87.$$

Example 18 (Reverse cash-and-carry arbitrage)
Suppose $S_0 = 100$, $r = 5\%$, $T = 1$, *and the market forward price is 102. The forward is too cheap relative to* 105.13. *A trader can:*

1. *short sell the stock and receive 100,*
2. *invest 100 at the risk-free rate,*
3. *enter a long forward at 102.*

At maturity, the investment grows to 105.13. Use 102 to buy the stock through the forward and return it to close the short sale. The locked-in profit is

$$105.13 - 102 = 3.13.$$

Option Bounds

No-arbitrage also gives inequalities. A European call on a non-dividend-paying stock cannot be worth more than the stock:

$$c \leq S_0.$$

It cannot be worth less than

$$S_0 - Ke^{-rT},$$

unless this number is negative. Therefore

$$c \geq \max(S_0 - Ke^{-rT}, 0).$$

Example 19 (Detecting a call-price violation)
Let $S_0 = 50$, $K = 45$, $r = 4\%$, *and* $T = 0.5$. *The lower bound is*

$$50 - 45e^{-0.04\times 0.5} \approx 50 - 44.11 = 5.89.$$

If the call trades for 4.80, it is too cheap relative to the bound. The quoted price violates no-arbitrage under the assumptions.

Put-Call Parity

For European options on a non-dividend-paying stock, put-call parity is

$$c + Ke^{-rT} = p + S_0.$$

The left side is a call plus a risk-free bond that pays K at maturity. The right side is a put plus the stock. At maturity both sides pay

$$\max(S_T, K).$$

Example 20 (Put-call parity)
Suppose $S_0 = 100$, $K = 100$, $r = 5\%$, $T = 1$, and a European call price is $c = 10.45$. Put-call parity implies

$$p = c + Ke^{-rT} - S_0 = 10.45 + 100e^{-0.05} - 100 \approx 5.57.$$

If the put trades at 7.00, the put is expensive relative to the call, stock, and bond combination.

> **Method.** No-arbitrage problems usually become easy when you build a payoff table. Write the payoff of each strategy in each future state. If two columns are identical, their current costs should match.

Probability

Random Variables

A random variable is a numerical outcome of an uncertain event. Before maturity, S_T is a random variable. After maturity, it is observed.

Example 21 (Random payoff)
A stock will be either 120 or 80 in one year. A call with strike 100 has payoff

$$(S_T - 100)^+.$$

State	S_T	*Call payoff*
Up	120	20
Down	80	0

The payoff is random because it depends on the future state.

Expected Value

The expected value is the probability-weighted average outcome. If X takes values $x_1, \ldots, x_n$ with probabilities $p_1, \ldots, p_n$, then

$$\mathbb{E}[X] = \sum_{i=1}^{n} p_i x_i.$$

Example 22 (Expected stock price)
A stock will be 120 with probability 0.6 and 80 with probability 0.4. Then

$$\mathbb{E}[S_T] = 0.6(120) + 0.4(80) = 104.$$

The expected price is not necessarily a price at which the stock can trade. It is a probability-weighted average.

Example 23 (Expected option payoff)
With the same states, a call with strike 100 pays 20 in the up state and 0 in the down state. Its expected payoff is

$$\mathbb{E}[(S_T - 100)^+] = 0.6(20) + 0.4(0) = 12.$$

This is not automatically the option price. Pricing also requires discounting and risk adjustment or replication.

Variance and Standard Deviation

Variance measures average squared deviation from the mean:

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Standard deviation is the square root of variance:

$$\sigma_X = \sqrt{\mathrm{Var}(X)}.$$

In finance, standard deviation is often called volatility.

Example 24 (Variance in a two-state model)
Let S_T be 120 with probability 0.6 and 80 with probability 0.4. We already computed $\mathbb{E}[S_T] = 104$. The variance is

$$\begin{aligned}\mathrm{Var}(S_T) &= 0.6(120 - 104)^2 + 0.4(80 - 104)^2 \\ &= 0.6(16)^2 + 0.4(-24)^2 \\ &= 384.\end{aligned}$$

The standard deviation is

$$\sqrt{384} \approx 19.60.$$

Covariance and Correlation

Covariance measures whether two variables move together:

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Correlation rescales covariance:

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Correlation always lies between -1 and 1.

Example 25 (Interpreting correlation)
If oil-price changes and airline stock returns have negative correlation, oil price increases tend to coincide with airline stock losses. This does not mean oil prices are the only cause of airline returns. Correlation is a co-movement measure, not a complete causal statement.

Example 26 (Covariance calculation)
Suppose two returns have the following equally likely outcomes.

State	R_A	R_B
1	0.10	0.06
2	0.02	0.01
3	−0.04	−0.02

The means are

$$\mathbb{E}[R_A] = \frac{0.10 + 0.02 - 0.04}{3} = 0.0267,$$
$$\mathbb{E}[R_B] = \frac{0.06 + 0.01 - 0.02}{3} = 0.0167.$$

The covariance is

$$\begin{aligned}\mathrm{Cov}(R_A, R_B) &= \frac{1}{3}\,[(0.10 - 0.0267)(0.06 - 0.0167) \\ &\quad + (0.02 - 0.0267)(0.01 - 0.0167) \\ &\quad + (-0.04 - 0.0267)(-0.02 - 0.0167)] \\ &\approx 0.00189.\end{aligned}$$

The covariance is positive because the returns move in the same direction across states.

Conditional Probability and Conditional Expectation

Conditional probability updates probabilities after information is observed. The notation

$$\mathbb{P}(A \mid B)$$

means the probability of event A given that event B has occurred.

Conditional expectation is the expected value after conditioning on information:

$$\mathbb{E}[X \mid \mathcal{I}].$$

In finance, $\mathcal{I}$ often represents current market information.

Example 27 (Conditional expectation)
A commodity price can be high or low next month. If a storage report is normal, the expected price is 70. If the report shows a shortage, the expected price is 82. Before seeing the report, the expectation averages over both possibilities. After seeing the report, the conditional expectation uses the relevant information set.

Independent Events and Independent Random Variables

Events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Random variables are independent if knowing one does not change the distribution of the other.

> **Warning.** Zero correlation is weaker than independence. If two variables are independent, their covariance is zero. But covariance can be zero even when the variables are not independent.

Normal and Lognormal Distributions

Normal Distribution

A normal random variable has the familiar bell-shaped distribution. We write

$$X \sim N(\mu, \sigma^2)$$

to mean that X is normally distributed with mean μ and variance σ^2.

The standard normal distribution is

$$Z \sim N(0, 1).$$

Any normal variable can be standardized:

$$Z = \frac{X - \mu}{\sigma}.$$

Example 28 (Standardization)
Suppose daily portfolio returns are approximately normal with mean 0.05% and standard deviation 1.20%. What is the standardized value of a return of -2.35%*?*

$$z = \frac{-0.0235 - 0.0005}{0.0120} = -2.$$

The return is two standard deviations below the mean.

Why Normal Distributions Appear in Finance

Normal distributions are useful because sums of many small shocks are often approximately normal. This is the central limit theorem intuition. In the Black–Scholes–Merton model, continuously compounded returns over a fixed horizon are normally distributed.

Interpretation. A normal distribution can take negative values. That is acceptable for returns, but not for stock prices. This is one reason the model uses normal log returns and lognormal prices.

Lognormal Distribution

A positive random variable S is lognormal if its logarithm is normal:

$$\ln S \sim N(\mu, \sigma^2).$$

Equivalently,

$$S = e^X$$

where X is normal.

Lognormal variables are always positive. This makes them natural for stock prices in the Black–Scholes–Merton model.

Example 29 (From normal log return to lognormal price)
Suppose

$$\ln(S_T/S_0) \sim N(0.04, 0.20^2).$$

Then S_T/S_0 is lognormal, and S_T is always positive. A one-standard deviation high log return is

$$0.04 + 0.20 = 0.24,$$

which corresponds to a gross return

$$e^{0.24} \approx 1.271.$$

That is a 27.1% simple return.

Simple Returns and Log Returns

The simple return from 0 to T is

$$R = \frac{S_T - S_0}{S_0}.$$

The log return is

$$r = \ln\left(\frac{S_T}{S_0}\right).$$

They are related by

$$r = \ln(1 + R), \qquad R = e^r - 1.$$

For small returns, $r \approx R$.

Example 30 (Simple and log return)
A stock rises from 50 to 55. The simple return is

$$R = \frac{55 - 50}{50} = 0.10.$$

The log return is

$$r = \ln(55/50) = \ln(1.10) \approx 0.0953.$$

For a 10% simple return, the log return is 9.53%.

Example 31 (Why log returns add)
A stock rises from 100 to 110 and then falls from 110 to 99. The two simple returns are 10% *and* -10%*, but the total simple return is* -1%*.*

The log returns are

$$\ln(110/100) = \ln(1.10), \qquad \ln(99/110) = \ln(0.90).$$

The total log return is

$$\ln(1.10) + \ln(0.90) = \ln(0.99),$$

which is exactly the log return from 100 to 99.

Calculus for Finance

Derivatives as Slopes

The derivative of $f(x)$ is the instantaneous rate of change of f with respect to x:

$$f'(x) = \frac{df}{dx}.$$

In finance, derivatives measure sensitivities. The option Greek Delta is

$$\Delta = \frac{\partial f}{\partial S},$$

the sensitivity of option value to the underlying price.

Example 32 (Derivative of a linear payoff)
For a long forward payoff

$$g(S_T) = S_T - K,$$

the derivative with respect to S_T *is*

$$g'(S_T) = 1.$$

The payoff changes one-for-one with the terminal spot price.

Example 33 (Derivative of a bond price with respect to yield)
A zero-coupon bond paying 100 at time T has price

$$B(y) = 100e^{-yT}.$$

The derivative with respect to yield is

$$\frac{dB}{dy} = -T100e^{-yT} = -TB(y).$$

The negative sign means that bond prices fall when yields rise.

Common Derivative Rules

$$\frac{d}{dx}x^n = nx^{n-1}, \qquad \frac{d}{dx}e^x = e^x, \qquad \frac{d}{dx}\ln x = \frac{1}{x}.$$

Product rule:

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).$$

Chain rule:

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x).$$

Example 34 (Chain rule in discounting)
Let

$$B(y) = e^{-yT}.$$

Set $u = -yT$. Then $B = e^u$ and $du/dy = -T$. The chain rule gives

$$\frac{dB}{dy} = e^u(-T) = -Te^{-yT}.$$

Partial Derivatives

When a function has several inputs, a partial derivative changes one input while holding the others fixed. If an option value is

$$f = f(S, t, \sigma, r),$$

then

$$\frac{\partial f}{\partial S}$$

changes S while holding t, σ, r fixed.

Example 35 (Partial derivatives)
Let

$$f(S, t) = S^2e^{-rt}.$$

Then

$$\frac{\partial f}{\partial S} = 2Se^{-rt}, \qquad \frac{\partial f}{\partial t} = -rS^2e^{-rt}.$$

The first derivative measures sensitivity to the price. The second measures sensitivity to time, holding S fixed.

Taylor Approximation

Taylor approximation replaces a nonlinear function with a local polynomial. First order:

$$f(x + \Delta x) \approx f(x) + f'(x)\Delta x.$$

Second order:

$$f(x + \Delta x) \approx f(x) + f'(x)\Delta x + \frac{1}{2}f''(x)(\Delta x)^2.$$

For an option value $f(S, t)$, the standard Greek approximation is

$$\Delta f \approx \Delta\Delta S + \frac{1}{2}\Gamma(\Delta S)^2 + \Theta\Delta t.$$

Here the first Δ on the right side is the Greek Delta, while ΔS is the change in the underlying price.

Example 36 (Delta-Gamma approximation)
An option has $\Delta = 0.60$ and $\Gamma = 0.04$. The stock price rises by 2. The Delta-Gamma approximation to the option-value change is

$$0.60(2) + \frac{1}{2}(0.04)(2)^2 = 1.20 + 0.08 = 1.28.$$

The Gamma term matters because the option value is curved in the stock price.

Optimization

Optimization chooses a variable to maximize or minimize an objective. In one dimension, an interior optimum usually satisfies the first-order condition

$$f'(x) = 0.$$

For a minimum, the second derivative should be positive:

$$f''(x) > 0.$$

Example 37 (Minimum-variance hedge ratio)
Let ΔS be the change in the spot price and ΔF the change in the futures price. A short hedge using hedge ratio h has change

$$\Delta V = \Delta S - h\Delta F.$$

The variance is

$$\mathrm{Var}(\Delta V) = \sigma_S^2 + h^2\sigma_F^2 - 2h\,\mathrm{Cov}(\Delta S, \Delta F).$$

Differentiate with respect to h:

$$\frac{d}{dh}\mathrm{Var}(\Delta V) = 2h\sigma_F^2 - 2\,\mathrm{Cov}(\Delta S, \Delta F).$$

Set this equal to zero:

$$h^* = \frac{\mathrm{Cov}(\Delta S, \Delta F)}{\sigma_F^2} = \rho \frac{\sigma_S}{\sigma_F}.$$

This is the minimum-variance hedge ratio.

Matrices and Portfolio Risk

Vectors

A vector is an ordered list of numbers. A portfolio with weights

$$w = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}$$

and asset returns

$$R = \begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ R_n \end{pmatrix}$$

has portfolio return

$$R_p = w^\top R = \sum_{i=1}^{n} w_i R_i.$$

Example 38 (Portfolio return)
A portfolio invests 40% in asset A and 60% in asset B. If returns are $R_A = 8\%$ and $R_B = 3\%$, then

$$R_p = 0.4(0.08) + 0.6(0.03) = 0.05.$$

The portfolio return is 5%.

Covariance Matrix

The covariance matrix collects variances and covariances:

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}.$$

For portfolio weights w, portfolio variance is

$$\sigma_p^2 = w^\top \Sigma w.$$

For two assets,

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \sigma_{12}.$$

Example 39 (Two-asset portfolio variance)
Asset A has volatility 20%, asset B has volatility 10%, and their correlation is 0.30. The portfolio weights are 50% and 50%. The covariance is

$$\sigma_{AB} = \rho\sigma_A\sigma_B = 0.30(0.20)(0.10) = 0.006.$$

The portfolio variance is

$$0.5^2(0.20)^2 + 0.5^2(0.10)^2 + 2(0.5)(0.5)(0.006) = 0.0155.$$

The portfolio volatility is

$$\sqrt{0.0155} \approx 0.1245.$$

So the portfolio volatility is about 12.45%.

Diversification

Diversification works when assets are not perfectly correlated. If two assets have correlation less than 1, the portfolio volatility can be below the weighted average of individual volatilities.

> **Derivative connection.** A hedge is a portfolio designed so that one position offsets risk in another. The effectiveness of the hedge depends on covariance, not only on the size of the two positions.

Statistics Used in the Course

Sample Mean and Sample Variance

Given observations $x_1, \ldots, x_n$, the sample mean is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

The sample variance is

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The denominator $n-1$ adjusts for the fact that the mean is estimated from the same data.

Example 40 (Sample volatility)
Suppose five daily returns are

$$0.4\%, \quad -0.6\%, \quad 1.0\%, \quad 0.2\%, \quad -0.5\%.$$

In decimals:

$$0.004, \quad -0.006, \quad 0.010, \quad 0.002, \quad -0.005.$$

The sample mean is

$$\bar{x} = 0.001.$$

The sample variance is

$$\begin{aligned} s^2 &= \frac{1}{4}\Big[(0.004-0.001)^2 + (-0.006-0.001)^2 \\ &\quad + (0.010-0.001)^2 + (0.002-0.001)^2 \\ &\quad + (-0.005-0.001)^2\Big] \\ &= 0.000044. \end{aligned}$$

The sample standard deviation is

$$s = \sqrt{0.000044} \approx 0.00663,$$

or 0.663% per day.

Annualizing Volatility

If daily returns are independent with constant variance, annual variance is approximately daily variance times the number of trading days. With 252 trading days,

$$\sigma_{\text{annual}} \approx \sigma_{\text{daily}}\sqrt{252}.$$

Example 41 (Annualized volatility)
If daily volatility is 1.2%, annualized volatility is

$$0.012\sqrt{252} \approx 0.1905.$$

So the annualized volatility is about 19.05%.

> **Assumption.** The square-root-of-time rule relies on weak dependence and roughly constant variance. It can be misleading during crises, for illiquid assets, or when volatility changes sharply.

Regression Beta

A simple regression of asset returns on market returns is

$$R_i = \alpha + \beta R_m + \varepsilon.$$

The slope is

$$\beta = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}.$$

Beta measures market exposure.

Example 42 (Equity index hedge)
A portfolio is worth USD 10 million and has beta 1.2 relative to an index. The index futures price is 5,000 and each futures contract is for 50 times the index. The notional per futures contract is

$$5{,}000 \times 50 = \$250{,}000.$$

The number of contracts for a short hedge is

$$N^* = \beta \frac{V_A}{V_F} = 1.2 \frac{10{,}000{,}000}{250{,}000} = 48.$$

The portfolio manager shorts 48 futures contracts to reduce market exposure.

Binomial Trees and Risk-Neutral Valuation

One-Step Binomial Model

In a one-step binomial model, the stock price moves from S_0 to either

$$S_u = S_0 u \qquad \text{or} \qquad S_d = S_0 d,$$

where $u > d$.

For a derivative with payoffs f_u and f_d, replication uses Δ shares and a bond position B:

$$\Delta S_0 u + Be^{rT} = f_u,$$

$$\Delta S_0 d + Be^{rT} = f_d.$$

Subtract the second equation from the first:

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d}.$$

Example 43 (One-step call by replication)
Let $S_0 = 100$, $u = 1.2$, $d = 0.8$, $K = 100$, $r = 5\%$, and $T = 1$. The stock prices are 120 and 80. The call payoffs are

$$f_u = 20, \qquad f_d = 0.$$

The hedge ratio is

$$\Delta = \frac{20 - 0}{120 - 80} = 0.5.$$

Solve for the bond position using the down state:

$$0.5(80) + Be^{0.05} = 0.$$

Thus

$$B = -40e^{-0.05} \approx -38.05.$$

The option value is

$$f_0 = \Delta S_0 + B = 0.5(100) - 38.05 = 11.95.$$

Risk-Neutral Probability

The same price can be written as a discounted expected payoff under the risk-neutral probability

$$p = \frac{e^{rT} - d}{u - d}.$$

Then

$$f_0 = e^{-rT}[pf_u + (1-p)f_d].$$

Example 44 (One-step call by risk-neutral probability)
Using the previous example,

$$p = \frac{e^{0.05} - 0.8}{1.2 - 0.8} \approx 0.6282.$$

The option price is

$$f_0 = e^{-0.05}[0.6282(20) + (1 - 0.6282)(0)] \approx 11.95.$$

This matches the replication price.

> **Interpretation.** Risk-neutral probabilities are pricing weights, not necessarily real-world probabilities. They are chosen so that discounted asset prices are consistent with no-arbitrage.

Backward Induction

In a multi-step tree, start at maturity and move backward one node at a time:

$$f = e^{-r\Delta t}[pf_u + (1-p)f_d].$$

For American options, compare continuation value with immediate exercise value at each node.

Example 45 (American put early exercise logic)
At a node, suppose an American put has strike 50 and the stock price is 42. Immediate exercise pays

$$50 - 42 = 8.$$

Suppose the discounted continuation value from holding the option is 7.40. The American value at the node is

$$\max(8, 7.40) = 8.$$

Exercise is optimal at that node in the tree.

Stochastic Processes

From Random Variables to Processes

A stochastic process is a collection of random variables indexed by time:

$$\{S_t : t \geq 0\}.$$

Instead of one uncertain future value, we model an uncertain path over time.

Example 46 (Daily price process)
A stock price observed every day can be written as

$$S_0, S_1, S_2, \ldots.$$

Before the future days arrive, $S_1, S_2, \ldots$ are random variables. After the days pass, they become observed values.

Markov Property

A process is Markov if the future depends on the present state, not on the full history:

$$\mathbb{P}(S_{t+\Delta t} \leq x \mid S_t, S_{t-1}, S_{t-2}, \ldots) = \mathbb{P}(S_{t+\Delta t} \leq x \mid S_t).$$

This is a simplifying assumption, not a universal truth.

> **Modeling warning.** The Markov property helps make pricing tractable. It does not mean real markets have no memory. Volatility clustering, liquidity stress, and institutional constraints often create path dependence.

Brownian Motion

A standard Brownian motion, or Wiener process, is usually written z_t. It has:

1. $z_0 = 0$,
2. independent increments,
3. normally distributed increments,
4. continuous paths.

Over a small time interval Δt,

$$\Delta z = \varepsilon\sqrt{\Delta t}, \qquad \varepsilon \sim N(0,1).$$

Thus

$$\mathbb{E}[\Delta z] = 0, \qquad \mathrm{Var}(\Delta z) = \Delta t.$$

Example 47 (Brownian increment)
If $\Delta t = 1/252$ year, then

$$\sqrt{\Delta t} = \sqrt{1/252} \approx 0.0630.$$

A one-standard-deviation Brownian increment over one trading day is about 0.0630 in annual time units.

Geometric Brownian Motion

The Black–Scholes–Merton model assumes the stock price follows

$$dS = \mu S\, dt + \sigma S\, dz.$$

This is geometric Brownian motion. The term $\mu S\, dt$ is drift. The term $\sigma S\, dz$ is random shock.

Dividing by S gives

$$\frac{dS}{S} = \mu\, dt + \sigma\, dz.$$

So μ is expected continuously compounded return per unit time and σ is volatility per square-root unit of time.

Example 48 (Interpreting the GBM shock)
Suppose annual volatility is $\sigma = 20\%$ and $\Delta t = 1/252$. A one-standard deviation daily price shock is approximately

$$\sigma\sqrt{\Delta t} = 0.20\sqrt{1/252} \approx 0.0126.$$

That is about 1.26% of the stock price.

Quadratic Variation Intuition

In ordinary calculus, terms like $(dt)^2$ are negligible. In stochastic calculus, Brownian increments satisfy

$$(dz)^2 = dt,$$

while

$$dt\, dz = 0, \qquad (dt)^2 = 0.$$

The informal reason is that dz is of order $\sqrt{dt}$, so $(dz)^2$ is of order dt.

> **Key idea.** The rule $(dz)^2 = dt$ is why Itô's lemma has an extra second-derivative term. That term becomes central in the Black–Scholes–Merton PDE.

Itô's Lemma

If

$$dx = a(x,t)\,dt + b(x,t)\,dz,$$

and $G = G(x,t)$, then Itô's lemma says

$$dG = \left(\frac{\partial G}{\partial x}a + \frac{\partial G}{\partial t} + \frac{1}{2}\frac{\partial^2 G}{\partial x^2}b^2\right)dt + \frac{\partial G}{\partial x}b\,dz.$$

Example 49 (Log of a GBM stock price)
Let

$$dS = \mu S\,dt + \sigma S\,dz$$

and define

$$G(S) = \ln S.$$

The derivatives are

$$G_S = \frac{1}{S}, \qquad G_{SS} = -\frac{1}{S^2}, \qquad G_t = 0.$$

Itô's lemma gives

$$\begin{aligned} d\ln S &= \left[\frac{1}{S}(\mu S) + \frac{1}{2}\left(-\frac{1}{S^2}\right)(\sigma^2 S^2)\right]dt + \frac{1}{S}(\sigma S)\,dz \\ &= \left(\mu - \frac{1}{2}\sigma^2\right)dt + \sigma\,dz. \end{aligned}$$

The log price has drift $\mu - \frac{1}{2}\sigma^2$, not μ.

Black–Scholes–Merton Building Blocks

Delta Hedging

If an option has value $f(S,t)$, its small change is approximately

$$df \approx f_S\,dS + f_t\,dt + \frac{1}{2}f_{SS}(dS)^2.$$

The option's exposure to the stock over a small move is f_S, called Delta. A portfolio that is long one option and short $\Delta = f_S$ shares has stock exposure approximately canceled over a very short interval.

Example 50 (Delta-neutral position)
An option has Delta 0.65. A trader is long 100 options, each written on one share. The stock exposure is approximately

$$100 \times 0.65 = 65$$

shares. To make the position delta-neutral, the trader shorts 65 shares.

Greeks as Sensitivities

Greek	Formula	Interpretation
Delta	$\Delta = \partial f / \partial S$	sensitivity to underlying price
Gamma	$\Gamma = \partial^2 f / \partial S^2$	curvature in underlying price
Theta	$\Theta = \partial f / \partial t$	sensitivity to calendar time
Vega	$\nu = \partial f / \partial \sigma$	sensitivity to volatility
Rho	$\rho = \partial f / \partial r$	sensitivity to interest rate

Example 51 (Greek-based scenario)
A portfolio has Delta $-20{,}000$, *Gamma 500, and Vega 75,000. The underlying price increases by 1 and implied volatility increases by 0.01. Ignoring time and interest-rate effects, the approximate value change is*

$$\begin{aligned} -20{,}000(1) + \frac{1}{2}(500)(1)^2 + 75{,}000(0.01) &= -20{,}000 + 250 + 750 \\ &= -19{,}000. \end{aligned}$$

Risk-Neutral Valuation

Risk-neutral valuation says that, under no-arbitrage, derivative prices can be computed as discounted expected payoffs under a special probability measure:

$$f_0 = e^{-rT}\mathbb{E}^{Q}[g(S_T)].$$

The superscript Q means the expectation is taken under risk-neutral probabilities.

For a stock under the risk-neutral measure,

$$dS = rS\,dt + \sigma S\,dz^{Q}$$

in the simplest no-dividend Black–Scholes–Merton model. The expected return changes from μ to r, but volatility remains σ.

> **Common confusion.** Risk-neutral valuation does not say investors are risk-neutral. It says derivative prices can be computed as if expected returns were the risk-free rate after risk has been correctly embedded in pricing probabilities.

Alternative Investment Mathematics

Internal Rate of Return

The internal rate of return is the discount rate that sets net present value to zero:

$$0 = \sum_{t=0}^{T} \frac{CF_t}{(1+\mathrm{IRR})^t}.$$

The initial investment is usually a negative cash flow.

Example 52 (IRR for a two-year investment)
An investment costs 100 today, pays 30 after one year, and pays 90 after two years. The IRR solves

$$-100 + \frac{30}{1+r} + \frac{90}{(1+r)^2} = 0.$$

Let $x = 1 + r$. Then

$$-100 + \frac{30}{x} + \frac{90}{x^2} = 0.$$

Multiply by x^2:

$$-100x^2 + 30x + 90 = 0.$$

Equivalently,

$$100x^2 - 30x - 90 = 0.$$

The positive solution is

$$x = \frac{30 + \sqrt{30^2 + 4(100)(90)}}{200} \approx 1.1105.$$

Thus

$$r \approx 0.1105.$$

The IRR is about 11.05%.

> **IRR warning.** IRR can be misleading when cash flows change signs more than once, when projects have different scale, or when interim cash flows cannot be reinvested at the IRR. NPV is usually the cleaner value measure.

Money Multiples

Private equity often reports multiples:

$$\text{TVPI} = \frac{\text{distributions} + \text{residual value}}{\text{paid-in capital}}.$$

DPI uses distributions only. RVPI uses residual value only.

Example 53 (Private equity multiples)
A fund has paid-in capital of 200, cumulative distributions of 90, and remaining net asset value of 170. Then

$$\text{DPI} = \frac{90}{200} = 0.45, \qquad \text{RVPI} = \frac{170}{200} = 0.85,$$

and

$$\text{TVPI} = 0.45 + 0.85 = 1.30.$$

The fund has generated total value equal to 1.30 times paid-in capital, but much of it remains unrealized.

Real Options

A real option is managerial flexibility with option-like payoff. Examples:

- option to delay investment,
- option to expand,
- option to abandon,
- option to switch inputs or outputs,
- option to stage financing.

Example 54 (Option to abandon)
A project has uncertain continuation value next year. If continued, it is worth either 150 or 60. The firm can abandon the project and sell assets for 80. The value next year is

$$\max(V, 80).$$

In the high state this is 150. In the low state this is 80, not 60. The abandonment option creates a floor, just like a put option creates downside protection.

Example 55 (Option to delay)
A firm can invest 100 today in a project whose value is uncertain next year. If the firm waits, it can invest only if the project value is high. Waiting has an opportunity cost because cash flows are delayed, but it also has option value because the firm avoids investing in bad states. The trade-off is the same logic as exercising versus holding an option.

Common Mistakes and Diagnostic Checks

Mistake 1: Mixing Percent and Decimal Units

Using 5 instead of 0.05 in an interest-rate formula changes the calculation from 5% to 500%.

Example 56 (Unit error)
The correct one-year growth factor at 5% is

$$e^{0.05} = 1.0513.$$

The incorrect calculation

$$e^{5} = 148.41$$

would imply a 14,741% return.

Mistake 2: Treating Expected Payoff as Price

The expected payoff is not generally the price. A derivative price must account for discounting, risk, and replication.

Example 57 (Expected payoff is not enough)
A risky payoff has expected value 100 in one year. If it is very risky, its price may be far below $100e^{-rT}$*. If it can be perfectly replicated, its price is determined by the replication cost, not by a subjective expected payoff.*

Mistake 3: Ignoring Contract Size

Futures and options quotes are often quoted per unit, but contracts cover many units.

Example 58 (Contract multiplier)
An index option quote changes by 3.20 index points. If the contract multiplier is 100, the dollar change per contract is

$$3.20 \times 100 = \$320.$$

For 25 contracts, the total change is

$$25 \times 320 = \$8{,}000.$$

Mistake 4: Confusing Real-World and Risk-Neutral Probabilities

Real-world probabilities describe statistical beliefs about future states. Risk-neutral probabilities are pricing weights that make no-arbitrage valuation work.

Example 59 (Two different probabilities)
In a one-step tree, the real-world probability of an up move may be 0.55. But the risk-neutral probability used for pricing may be 0.48. The derivative price uses 0.48 in the no-arbitrage formula, not because 0.48 is the true forecast, but because it is the pricing weight consistent with the stock and bond prices.

Worked Review Problems

Example 60 (Forward price with income)
A stock is worth 80 today and will pay a known dividend with present value 2 before maturity. The continuously compounded risk-free rate is 4% and maturity is one year. The forward price is

$$F_0 = (S_0 - I)e^{rT} = (80 - 2)e^{0.04} \approx 81.18.$$

The dividend lowers the forward price because the forward buyer does not receive the dividend before delivery.

Example 61 (Put-call parity arbitrage direction)
Let $S_0 = 40$, $K = 40$, $r = 5\%$, $T = 1$, $c = 4.50$, and $p = 1.00$. Put-call parity requires

$$c + Ke^{-rT} = p + S_0.$$

The left side is

$$4.50 + 40e^{-0.05} \approx 42.55.$$

The right side is

$$1.00 + 40 = 41.00.$$

The left side is expensive. Sell the left side and buy the right side: short the call, borrow the present value of 40 by shorting the bond, buy the put, and buy the stock. This locks in the price difference under the assumptions of European exercise, no dividends, and frictionless trading.

Example 62 (Minimum-variance hedge)
A firm has exposure to a spot commodity. The monthly standard deviation of spot price changes is 3.0, the monthly standard deviation of futures price changes is 2.5, and the correlation is 0.80. The optimal hedge ratio is

$$h^* = \rho \frac{\sigma_S}{\sigma_F} = 0.80 \frac{3.0}{2.5} = 0.96.$$

If the firm wants to hedge 100,000 units and each futures contract is for 10,000 units, the number of contracts is

$$N^* = 0.96 \frac{100{,}000}{10{,}000} = 9.6.$$

In practice the firm must choose 9 or 10 contracts, then consider transaction costs, liquidity, and basis risk.

Example 63 (Discounted expected payoff under risk-neutral probability)
A derivative pays 30 in the up state and 0 in the down state after one year. The risk-neutral probability of the up state is 0.40 and the risk-free rate is 5% continuously compounded. The price is

$$e^{-0.05}[0.40(30) + 0.60(0)] = 11.41.$$

Example 64 (Delta-Gamma-Vega approximation)
A trader owns an option portfolio with Delta 8,000, Gamma -300, and Vega 45,000. The underlying falls by 1.5 and implied volatility rises by 0.02. Ignoring Theta and Rho, the approximate change is

$$\begin{aligned} 8{,}000(-1.5) + \frac{1}{2}(-300)(-1.5)^2 + 45{,}000(0.02) &= -12{,}000 - 337.5 + 900 \\ &= -11{,}437.5. \end{aligned}$$

Checklist for Students

After reading this note, you should be able to:

- convert percentages, decimals, and basis points;
- use annual, periodic, and continuous compounding;
- discount a single cash flow and a stream of cash flows;
- compute zero rates, discount factors, and simple forward rates;
- read payoff notation such as $(S_T - K)^+$;
- distinguish payoff, profit, and value;
- build payoff tables for forwards, calls, puts, and combinations;
- state the law of one price and use it in no-arbitrage arguments;
- compute expectation, variance, covariance, and correlation;
- distinguish simple returns and log returns;
- explain why lognormal prices are positive;
- compute basic derivatives and partial derivatives;
- use Taylor approximations to interpret Greeks;
- derive the minimum-variance hedge ratio;
- compute portfolio variance from variances and covariance;
- annualize volatility using the square-root-of-time rule;
- price a one-step binomial option by replication and risk-neutral probability;
- explain the intuition behind Brownian motion, GBM, and Itô's lemma;
- compute IRR and money multiples for simple alternative investment examples.

Practice Questions

1. Convert 125 basis points into a percentage and a decimal.
2. A cash flow of 10,000 arrives in 18 months. The continuously compounded risk-free rate is 3.5%. Compute its present value.

3. An investment grows from 75 to 90 over two years. Compute the simple two-year return and the continuously compounded annual return.

4. A long forward has delivery price 62. Compute the payoff for $S_T = 45, 62, 70, 88$.

5. A call has strike 100 and premium 6. Compute payoff and profit for $S_T = 80, 100, 106, 125$, ignoring financing.

6. A put has strike 50 and premium 3. Compute payoff and profit for $S_T = 35, 47, 50, 60$, ignoring financing.

7. Explain in words why a call option payoff is convex.

8. A non-dividend-paying stock has $S_0 = 120$, $r = 4\%$, and $T = 0.75$. Compute the no-arbitrage forward price using continuous compounding.

9. If the market forward price in the previous question is 128, describe the arbitrage strategy and compute the locked-in profit.

10. A European call has $S_0 = 75$, $K = 70$, $r = 3\%$, $T = 1$, and price 3.50. Does it violate the lower bound? Show the calculation.

11. Given $S_0 = 100$, $K = 105$, $r = 5\%$, $T = 1$, and $c = 8$, use put-call parity to compute the European put price.

12. A random variable is 10 with probability 0.25 and 20 with probability 0.75. Compute its expectation and variance.

13. Asset A has volatility 25%, asset B has volatility 15%, and their correlation is -0.20. Compute their covariance.

14. A portfolio has weights 30% and 70% in the two assets from the previous question. Compute portfolio variance and volatility.

15. Daily volatility is 0.9%. Annualize it using 252 trading days.

16. A regression beta is defined as $\beta = \mathrm{Cov}(R_i, R_m)/\mathrm{Var}(R_m)$. If $\mathrm{Cov}(R_i, R_m) = 0.0012$ and $\mathrm{Var}(R_m) = 0.0008$, compute beta.

17. In a one-step binomial tree, $S_0 = 50$, $u = 1.3$, $d = 0.8$, $K = 55$, $r = 4\%$, and $T = 1$. Price the European call by risk-neutral valuation.

18. For the previous question, compute the replicating Delta.

19. An option has Delta 0.40 and Gamma 0.03. Approximate the price change when the underlying rises by 5, ignoring time and volatility changes.

20. A stock follows GBM with annual volatility 30%. What is the approximate one-standard-deviation daily percentage shock using 252 trading days?

21. Use Itô's lemma to explain why $d \ln S$ has drift $\mu - \frac{1}{2}\sigma^2$ when $dS = \mu S dt + \sigma S dz$.

22. A private equity fund has paid-in capital of 500, distributions of 220, and residual value of 430. Compute DPI, RVPI, and TVPI.

23. A project costs 1,000 and pays 400, 450, and 500 over the next three years. Compute NPV at a 9% discount rate.

24. Give one example of a real option and map it to a financial option payoff.